Bengali Word Sense Disambiguation Using Supervised approach (BERT)

by

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Under the supervision of

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We declare that this capstone project report entitled *Bengali Word Sense Disambiguation Using Supervised approach (BERT)* is the result of our own work except as cited in the references. The capstone project report has not been accepted for any degree and is not concurrently submitted in candidature of any other degree.

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To my beloved elder sister
whose unwavering support, endless encouragement, and quiet strength have been a guiding light in my life and throughout this journey. Your faith in me has been my greatest motivation. This work is as much yours as it is mine.

– Ariful Islam Bhoiyan

ABSTRACT

Word Sense Disambiguation (WSD) is a pivotal challenge in Natural Language Processing (NLP) for morphologically complex languages like Bengali, spoken by over 250 million people. Polysemous Bengali words, with multiple meanings based on context, create significant obstacles for applications such as machine translation and information retrieval. This study introduces a supervised WSD framework utilizing transformer-based BERT models to tackle these issues. We present a curated Bengali WSD dataset with contextually labeled examples, encompassing a variety of polysemous words and their senses. Two BERT-based models—csebuetnlp/BanglaBERT, sagorsarker/bangla-bert-base, and a multilingual variant—are evaluated within a consistent pipeline. The dataset is divided into 80% for training and 20% for testing (Validation 10% and testing 10%), with preprocessing to address Bengali's morphological intricacies. Model performance is assessed using accuracy, precision, recall, F1-score, and confusion matrix analysis. The csebuetnlp/BanglaBERT model achieved the highest performance, with 88.60% accuracy, 88.93% precision, 89.82% recall, and 89.38% F1-score, surpassing sagorsarker bangla-bert-base (83.12% accuracy). Class-wise analysis highlights robust performance in semantically distinct categories but reveals challenges with nuanced or underrepresented senses.Given its performance in terms of accuracy, stability, and generalization, the csebuetnlp/BanglaBERT model emerges as a promising option for Bengali Word Sense Disambiguation. The accompanying framework and dataset contribute significantly to progress in Bengali NLP, particularly in low-resource, high-complexity scenarios.

Keywords: Word Sense Disambiguation, Bengali NLP, Transformer Models, BanglaBERT, Polysemy, Supervised Learning, Contextual Embeddings

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Chapter 1

Introduction

Natural Language Processing (NLP) involves the interaction between computers and human languages, with Word Sense Disambiguation (WSD) being a fundamental problem. It involves identifying the correct sense of a word based on its context in a sentence, crucial for machine translation, information retrieval, question answering, and other downstream applications [1]. Accurately determining these meanings is essential for accurate translation and comprehension. In the context of Bengali, a morphologically rich and syntactically complex language spoken by more than 250 million people [2], WSD remains a relatively underexplored area. Many Bengali words are polysemous, meaning they have multiple meanings that vary depending on context. Incorrectly interpreting such words may cause serious misunderstandings in translation, dialogue systems, or search engines. Traditional WSD approaches, such as dictionary-based, knowledge-based, or supervised classification with hand-engineered features, have limited success in Bengali due to the following. Lack of comprehensive sense inventories like WordNet (available in English). Scarcity of annotated corpora. High inflection and derivation in Bengali morphology. Variability in syntactic structures across dialects and registers. Recent advances in transformer-based models, particularly BERT, have revolutionized Bengali NLP by enabling models to learn contextual embeddings, making them ideal for tasks like Word Sight Decoding. Variants such as **sagorsarker/bangla-bert-base** [3], **csebuetnlp/banglabert** [14], and multilingual models have opened new possibilities. However, applying BERT-based models to Bengali WSD presents unique challenges: A lack of benchmark WSD datasets with contextually labeled examples. Need for domain-general examples to ensure models can generalize

across topics. Performance evaluation across different transformer models in a consistent pipeline.

This study proposes a supervised WSD framework for Bengali that leverages BERT-based transformer architectures. It introduces a curated Bengali WSD dataset and compares the performance of two models sagorsarker/bangla-bert-base,csebuetnlp/banglaber. This task using accuracy, F1-score, and confusion matrix-based analysis.

1.1 Problem statement

Despite the growing availability of transformer-based models for Bengali, the task of Word Sense Disambiguation remains largely unsolved. Most existing research is either limited to English or restricted to shallow approaches in Bengali (e.g., rule-based or statistical). A lack of large-scale, annotated datasets has also inhibited experimentation with deep learning methods for Bengali WSD.

The central research problem addressed in this study is:

"How do different BERT-based models perform on annotated Bengali datasets for Word Sense Disambiguation in a supervised setting?"

This problem encapsulates several sub-problems:

Can we construct a labeled WSD dataset for Bengali that covers diverse words and senses?

"How effectively do BERT-based models handle contextual ambiguity, particularly when distinguishing between closely related senses?"

What are the limitations of current models in low-resource, high-complexity language settings like?

1.2 Aims, objectives and Motivation

1.2.1 Aims

The primary objective of this study is to develop a supervised learning-based Word Sense Disambiguation (WSD) system for the Bengali language by leveraging transformer-based models, particularly BERT and its variants. This research contributes to the Bengali NLP landscape by constructing a high-quality, annotated

WSD dataset and benchmarking state-of-the-art models on this resource. The ultimate goal is to enable machines to accurately interpret the nuanced meanings of polysemous Bengali words across diverse contexts—a capability that is crucial for a wide range of downstream applications, including question answering, information retrieval, and machine translation. To achieve this overarching objective, the study outlines the following specific aims:

- **Design and Develop a Supervised WSD System:** Using refined transformer models, formulate Bengali WSD as a multi-class classification issue and develop a system that can correctly disambiguate polysemous words based on sentence-level context.
- **Construct a Rich Annotated Bengali WSD Dataset:** The dataset will comprise at least 120 polysemous Bengali words, each annotated with three to five distinct senses. For every sense, 40 to 50 contextually diverse example sentences will be manually validated. To ensure broad linguistic coverage, samples will be collected from various sources, including books, online articles, news reports, social media posts, and conversational texts. The dataset will be formatted in a machine-readable structure, such as JSON, to enable seamless integration with contemporary NLP pipelines.
- **Utilize Pre-trained Bengali Transformer Models:** Employ and fine-tune three prominent pre-trained transformer models — `sagorsarker/bangla-bert-base`, `csebuetnlp/banglabert` — to evaluate their effectiveness in capturing semantic nuances in Bengali for WSD tasks. Leverage the Hugging Face Transformers library and PyTorch as the technical backbone.
- **Incorporate Strategies for Class Imbalance:** The intrinsic class imbalance caused by varying sense frequency distributions in natural language corpora can adversely affect model performance. To mitigate this, techniques such as stratified sampling are commonly employed. In this study, we aim to balance the dataset by ensuring an equal number of examples per sense label, thereby addressing the imbalance and promoting more robust model training.
- **Implement Comprehensive Evaluation Metrics:** Utilize the Accuracy, Precision, Recall, F1-Score, and confusion matrices to assess each model. Monitor

learning curves (accuracy and loss per epoch) to spot problems with generalization and overfitting. Analyze qualitative errors to find systemic flaws in the model.

- **Enable Model Comparison and Insightful Analysis:** Examine how well the chosen models perform in various language categories and syntactic complexity levels. Examine their capacity to generalize across unclear structures or unrecognized word senses, and note the advantages and disadvantages of each model.

1.2.2 Objective

To achieve the aim of this research, the following objectives are set:

1. **Data Construction**
    - Collect polysemous Bengali words and define their distinct senses.
    - Manually annotate at least 40-120 words, each with 4-5 senses, and 40-50 context-rich sentences per sense.
    - Store the dataset in a structured format (e.g., JSON) suitable for transformer input pipelines.
2. **Model Selection and Fine-tuning**
    - Choose transformer models pretrained on Bengali:
        - `sagorsarker/bangla-bert-base`
        - `csebuetnlp/banglabert`
    - Fine-tune each model using the training set while applying stratified validation/test splits.
3. **Evaluation**
    - Evaluate model performance using Accuracy, Precision, Recall, and F1-Score.
    - Generate and interpret confusion matrices to analyze sense-level misclassifications.

- Visualize training curves (loss and accuracy) and identify signs of overfitting or underfitting.

4. **Comparison and Analysis**
    - Compare multiple models using various performance metrics, and analyze their respective strengths and weaknesses.
    - Perform error analysis to identify common misclassification patterns and uncover any systematic biases present in the models.
    - Determine if any model shows better generalization across different sense categories.
5. **Documentation and Publication**
    - Make the dataset and code publicly available in a reproducible format to support future research.

1.2.3 Motivation

The motivation behind this research is multifaceted:

- **Lack of Bengali NLP Resources:**
  The majority of Word Sense Disambiguation (WSD) systems are designed for European languages, including English. Because Bengali is a low-resource language, benchmarking and model improvement are challenging due to the lack of datasets and resources for supervised WSD tasks [5].
- **Utility in Downstream Tasks:**
  For idiomatic idioms and domain-specific terminology in particular, accurate sense disambiguation is essential to improving downstream applications like machine translation. Additionally, it enhances Bengali voice assistant and chatbot interactions by giving more accurate and contextually relevant answers [6].
- **Transformer Potential:**
  Transformer-based models, such as BERT, are intrinsically ideal for WSD owing to their extensive contextual awareness of language. Their demonstrated success in other languages drives their applicability in Bengali [4].

1.3 Project Specification

This project is titled *Supervised Bengali Word Sense Disambiguation Using BERT-based Models* and is framed as a supervised classification task. The input consists of Bengali sentences containing polysemous words, and the output is a predicted sense label. The key technical components include:

- A bespoke Bengali WSD dataset including at least 122 polysemous words, each with 3-5 senses and around 50 annotated example sentences per sense.
- A modeling pipeline created with the Hugging Face Transformers library and a PyTorch backend.
- Evaluation through confusion matrices, classification reports, and training plots to analyze performance and model behavior.
- Cross-model evaluation using three transformer-based models: `sagorsarker/bangla-bert-base`, `csebuetnlp/ banglabert`.

Chapter 2

Literature Review

Word Sense Disambiguation (WSD) is a core task in Natural Language Processing (NLP) that involves identifying the correct meaning of a word based on its contextual usage. In recent years, deep learning (DL) techniques—particularly transformer-based models—have significantly improved WSD accuracy across many languages, including Bengali. Numerous studies have explored diverse methodologies for WSD, each presenting its own strengths and limitations.

In 2022, introduced a benchmark dataset tailored for Bengali WSD, comprising 100 polysemous words, each annotated with 3 to 4 distinct senses. Although the dataset is relatively small and limited in scope, it provides crucial insights into the unique challenges of Bengali WSD and serves as a foundation for evaluating and benchmarking future models [7].

Pal and Saha (2019) made a notable contribution by proposing an unsupervised approach that combined Principal Component Analysis (PCA) with Bengali WordNet to perform WSD. Their method achieved an accuracy of 61%, reflecting the potential of classical techniques in low-resource settings. However, the absence of supervised learning and transformer-based models in their framework limited the effectiveness of contextual understanding compared to more modern architectures such as BERT [5].

Further demonstrating the effectiveness of hybrid deep learning techniques, Saidi and Jarray (2023) proposed a BERT-CNN model for Arabic WSD. By integrating transformer-based representations with convolutional neural networks, they achieved an impressive accuracy of 96.42%. Their work, including variants like ArabicBERT-CNN and AraBERT-CNN, showcased the advantages of combining

contextual embeddings from pre-trained models with task-specific neural structures [8].

In 2024, Holanda do Nascimento et al. focused on WSD for Portuguese using BERTimbau, a transformer model pretrained specifically for the Portuguese language. Their system achieved an accuracy of 84%, underscoring the utility of language-specific BERT variants in multilingual WSD. Despite its effectiveness, the model encountered difficulties in disambiguating polysemous terms in low-resource scenarios, indicating the need for further fine-tuning and data augmentation [9].

Same year, Kumari and Lobiyal explored WSD in Hindi using distributed word representations, including Word2Vec (Skip-Gram) and Probabilistic Latent Semantic Analysis (PLSA). Their approach reported accuracies of 74.12% and 72%, respectively. However, their exclusion of transformer-based models highlighted a significant limitation, as contemporary architectures like BERT have proven superior in handling semantic ambiguity and context-dependent word meanings [10].

In 2020, a monolingual Bangla-BERT model was introduced to address the limitations of multilingual models like mBERT, which struggle with resource-constrained languages such as Bengali due to limited training data and shared weights across many languages. This model was pre-trained on BanglaLM, a large-scale corpus of 40 GB of Bengali text, making it the largest Bangla language model dataset to date. Bangla-BERT significantly outperformed multilingual BERT and prior approaches across various NLP tasks, including binary linguistic classification, multilabel extraction, and named entity recognition (NER). It also demonstrated superior performance compared to non-contextual models such as fastText and Word2Vec. Furthermore, through transfer learning combined with hybrid deep learning architectures like LSTM, CNN, and CRF for NER, Bangla-BERT consistently surpassed state-of-the-art results on benchmark datasets including Banfakenews, Sentiment Analysis on Bengali News Comments, and Cross-lingual Sentiment Analysis in Bengali, achieving improvements of 3.52%, 2.2%, and 5.3% respectively [4].

In 2021 Sagar Sarker introduced the Bangla BERT model, marking one of the earliest efforts to develop a transformer-based architecture tailored to Bengali. Although constrained by the size and scope of the training data, this model demonstrated the feasibility of leveraging transformers for Bengali NLP tasks. Bangla BERT laid the groundwork for future advancements in the field and is expected to

play a central role in addressing linguistic challenges in resource-scarce settings like Bengali [3].

Title	Publication Date	Author	Model	Accuracy
Word Sense Disambiguation in Bengali language using unsupervised methodology with modifications	27 June 2019	Alok Ranjan Pal & Diganta Saha	PCA(Principal Component Analysis) + Bengali WordNet	61%
Stacking of BERT and CNN Models for Arabic Word Sense Disambiguation	18 November 2023	Rakia Saidi, Fethi Jarray	3BERT-CNN (proposed), ArabicBERT-CNN, AraBERT-CNN,mBERT-CNN	96.42%, 95.07%, 93.18%, 94.97%
A Word Sense Disambiguation Method Applied to Natural Language Processing for the Portuguese Language	22 February 2024	Clovis Holanda do Nascimento, Vinicius Cardoso Garcia; Ricardo de Andrade Araújo	BERTimbau	84%

Figure 2.1: Comparison of different BERT models used in the WSD task

Chapter 3

Methodology

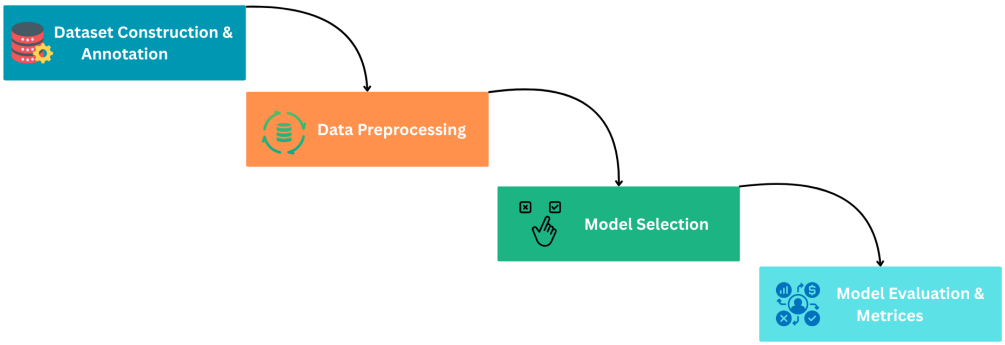


Figure 3.1: Methodology

3.1 Dataset Construction and Annotation:

The dataset for Bengali Word Sense Disambiguation (WSD) was constructed using a diverse range of Bengali textual sources to ensure rich and varied language use. We collected data from multiple online resources, including *online archives, articles, social media posts, books, and news articles*. These sources helped capture different registers and contexts in which polysemous Bengali words are used.

Initially, we stored data *word by word* in individual JSON files. Each JSON file contained:

- The **target Bengali word**
- Its **possible senses**
- Multiple **example sentences** for each sense

To ensure linguistic accuracy, **native Bengali linguists** manually annotated the sentences, assigning the correct sense of the target word within each context. The annotation process was carefully reviewed to maintain high-quality standards.

After preparing individual JSON files for each word, we *merged all 122 words* into a **single JSON file** to form the final dataset. The resulting dataset contains **over 22,626 annotated sentences**, with at least **3 senses per word**. Each entry in the JSON format includes the following fields:

- `"word"` – the target Bengali word
- `"sense"` – the sense label
- `"examples"` – a list of example sentences illustrating each sense

This structured and comprehensive dataset serves as the foundation for training and evaluating machine learning models for Bengali WSD tasks.

	Numbers
Words	122
Senses	440
Examples	22626

Table 3.1: Amount of words, senses, and examples.

3.2 Data Preprocessing

Data preprocessing is pivotal in any Natural Language Processing (NLP) task, especially for languages like Bengali, where textual data often comes from diverse and noisy sources. Proper preprocessing ensures that the raw text is transformed into a clean, consistent, and meaningful format, allowing the model to focus on learning

true semantic patterns rather than irrelevant noise. In the context of Word Sense Disambiguation (WSD), where subtle contextual differences determine meaning, unprocessed data can introduce ambiguity, bias, and inconsistencies that severely degrade model performance. Without adequate preprocessing, models are more likely to overfit to noise, misinterpret context, and exhibit poor generalization on unseen data. Moreover, imbalanced class distributions and encoding inconsistencies can lead to skewed predictions, unfair evaluations, and unstable training dynamics. By carefully cleaning, normalizing, balancing, and analyzing the dataset, we ensure the model is trained on high-quality input that accurately reflects real-world Bengali usage's linguistic richness and variability.

3.2.1 Normalization and Cleaning

Text normalization is a critical step in preparing Bengali text for modeling, particularly due to the language's rich diversity, informal usage, and regional variations. One of the primary challenges in processing Bengali is the presence of encoding inconsistencies, where characters that appear similar may be represented differently at the byte level. To address this, we applied Unicode normalization using the NFC (Normalization Form C) standard to ensure uniform character representation. We also removed non-standard characters, including extra spaces, informal symbols, and special characters such as emojis and irregular punctuation, to eliminate noise from the data. This normalization process was accomplished through a combination of manual inspection and automated Python scripts, ensuring both precision and scalability. The manual review helped identify edge cases and language-specific anomalies, while the scripts efficiently handled large-scale cleaning. These efforts ensured that the input data was clean, consistent, and semantically reliable, enabling the model to focus on meaningful patterns. Without such normalization, the model could misinterpret encoded forms, become biased by noisy artifacts, and yield unstable or inaccurate predictions.

3.2.2 Handling Imbalanced Data:

Handling imbalanced data is essential in any classification task, especially in Word Sense Disambiguation (WSD), where the distribution of senses across words can vary significantly. If the dataset is heavily skewed toward certain word senses, the

model may become biased, learning to predict the majority class more often while ignoring less frequent, but equally important, senses. This leads to poor generalization, low accuracy for minority classes, and misleading evaluation metrics. To mitigate this issue, we aimed to maintain a balanced dataset by ensuring that each sense of a word had approximately the same number of example sentences. By keeping the data size consistent across all senses, we ensured that the model received equal exposure to each category, which promotes fair learning and improves its ability to disambiguate words accurately across diverse contexts. This balance was achieved through a combination of oversampling underrepresented senses and limiting the number of examples for overrepresented ones, resulting in a more uniform and effective training set.

3.2.3 Tokenization and Input Formatting:

Tokenization is a crucial step in converting raw text into a format suitable for BERT-based models:

- **Padding and Truncation:**
  Sentences that are too long are truncated, and sentences that are too short are padded with special tokens ([PAD]). This ensures that all input sentences are of equal length, matching the model's input size.

- **Attention Masks:**
  Attention masks are generated to distinguish between real tokens and padding tokens. This is critical, as padding tokens should not be attended to during training or inference.

- **Special Tokens:**
  We use custom special tokens "TGT" and "/TGT" placed before and after the target word to explicitly highlight it in each sentence. This helps the model focus its attention on the specific word that requires disambiguation, improving its ability to understand the relevant context and accurately determine the word's correct sense.

3.2.4 Stratified Train-Test Split:

To ensure that each sense of the target word is equally represented in both the training and testing datasets, a stratified train-test split is performed:

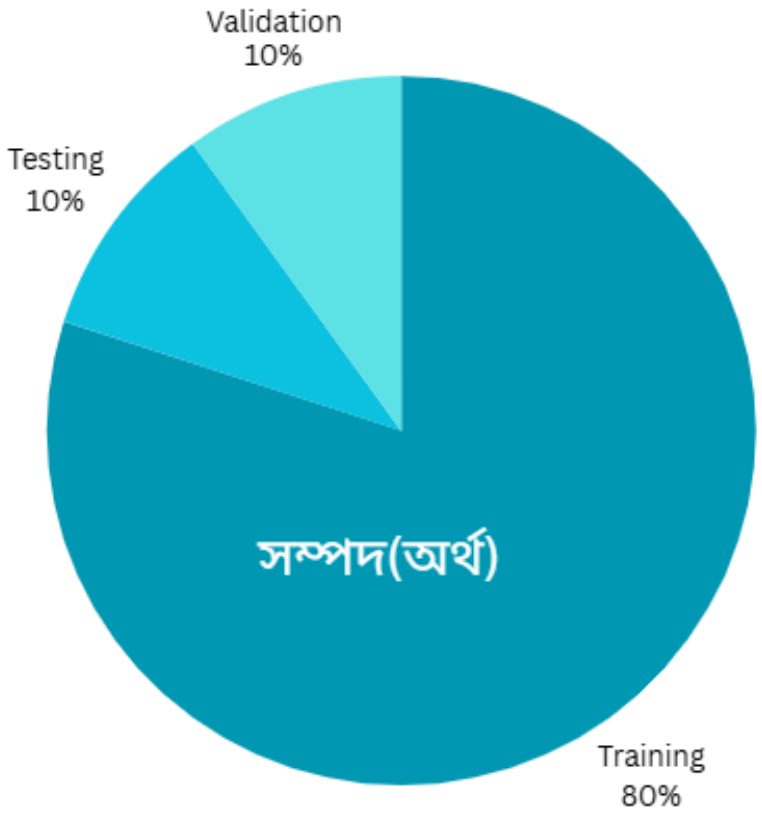


Figure 3.2: Visualization of a word sense data splitting

- **80% for Training, 10% for Testing and 10% for Validation:**
  To ensure effective model training and reliable evaluation, we divided the data set into three subsets: 80% for training, 10% for validation, and 10% for testing. This split allows the model to learn from a substantial portion of the data while reserving separate, unseen samples for both validation and final evaluation. The training set is used to optimize model parameters, the validation set helps tune hyperparameters and monitor overfit during training, and the test set provides an unbiased measure of model performance.

- **Stratification:**
  The split is done in such a way that each sense is proportionally represented in both the training and test sets. This ensures that no word sense is underrepresented or overrepresented in the test data, providing a fair evaluation of model performance.

3.3 Model Selection

Our project focuses on Word Sense Disambiguation (WSD) in Bangla, where the goal is to identify the correct meaning of a word based on its context in a sentence. To tackle this, we chose the BERT-base Transformer model because of its strong ability to understand context and semantic relationships—key aspects when dealing with ambiguous words.

Unlike traditional models like LSTMs, which process text in one direction and often struggle with long-range dependencies, BERT looks at both the left and right context of each word. This bidirectional approach helps the model build a much deeper understanding of how words relate to one another within a sentence [11].

BERT's contextual word embeddings are also much more accurate than static embeddings when it comes to capturing word meanings in different contexts. These embeddings have been shown to align closely with how humans distinguish between senses of a word [12]. Transformer-based models like BERT have also consistently outperformed older approaches in WSD tasks, making them a solid choice for this type of problem [13].

3.3.1 Fine-Tuning Pretrained Transformer Models:

For the WSD task, two different pretrained BERT-based models are fine-tuned:

- **Bangla BERT (sagorsarker/bangla-bert-base):** This is a BERT model specifically pretrained on Bengali text. It captures the syntactic and semantic structure of the Bengali language, making it suitable for tasks like word sense disambiguation.

- **BanglaBERT (csebuetnlp/banglabert):** Another BERT variant trained on a substantial 27.5 GB corpus of Bengali text, known as Bangla2B+—collected from 110 popular websites. The model effectively captures the syntactic and semantic nuances of Bangla and outperforms other models on Bangla NLP benchmarks, making it ideal for Word Sense Disambiguation **?**.

3.3.2 Model Training and Optimization:

Training the models involves the following hyperparameters:

- **Learning Rate:** A learning rate of 3e-5 is used. This is a typical starting point for fine-tuning BERT-based models, and it has shown good performance across multiple NLP tasks.
- **Batch Size:** The batch size is set to 16 to prevent memory overload while maintaining enough examples per update to stabilize training.
- **Epochs:** The models are trained for 15 epochs. Early stopping is applied if the validation loss stops improving after 3 consecutive epochs, preventing overfitting.
- **Optimizer:** The Adam optimizer is used with weight decay for regularization. This optimizer is known for its performance with BERT-based models.

3.4 Model Evaluation and Metrics

Model evaluation is crucial for determining the efficacy of fine-tuned models in the Word Sense Disambiguation (WSD) task. The model's performance is evaluated using two important metrics: accuracy and F1 score. Accuracy measures how many right predictions were made, but it does not take into consideration the class imbalance that is common in WSD assignments. The F1 score, on the other hand, is especially useful since it balances accuracy and recall, providing a more nuanced assessment when courses are unevenly distributed. A higher F1 score shows that the model has trained to correctly detect both positive and negative events for each sense.

Additionally, misclassifications are identified using confusion matrices created for each model. In these matrices, the rows represent genuine word senses and the columns represent anticipated word senses. The diagonal members of the matrix represent valid predictions, but the off-diagonal elements reflect where the model made mistakes. To make the confusion matrices more understandable, heatmaps are utilized to graphically show the frequency of misclassifications. By evaluating these heatmaps, it is possible to identify certain senses or classes in which the model suffers, offering useful information for future improvements, such as model refinement or dataset expansion.

3.4.1 Accuracy and F1 Score

Model performance is evaluated using accuracy and F1 score:

- **Accuracy:** Measures the percentage of correctly classified instances across all word senses. It gives an overall view of how well the model performs but doesn't take into account class imbalances.
- **F1 Score:** Measures the balance between precision (correct positive predictions) and recall (all true positives) for each word sense. The F1 score is especially useful for evaluating models when classes are imbalanced, as it provides a harmonic mean of precision and recall.

3.4.2 Confusion Matrix

A confusion matrix is generated for each model to visualize the misclassifications:

- Rows represent the true word senses, while columns represent the predicted word senses.
- The diagonal of the matrix shows the correct predictions, while off-diagonal elements indicate misclassifications.

Heatmaps of confusion matrices provide a clear visual understanding of where the model is making mistakes, allowing for targeted model improvements (e.g., through data augmentation or tuning).

3.5 Timeline:

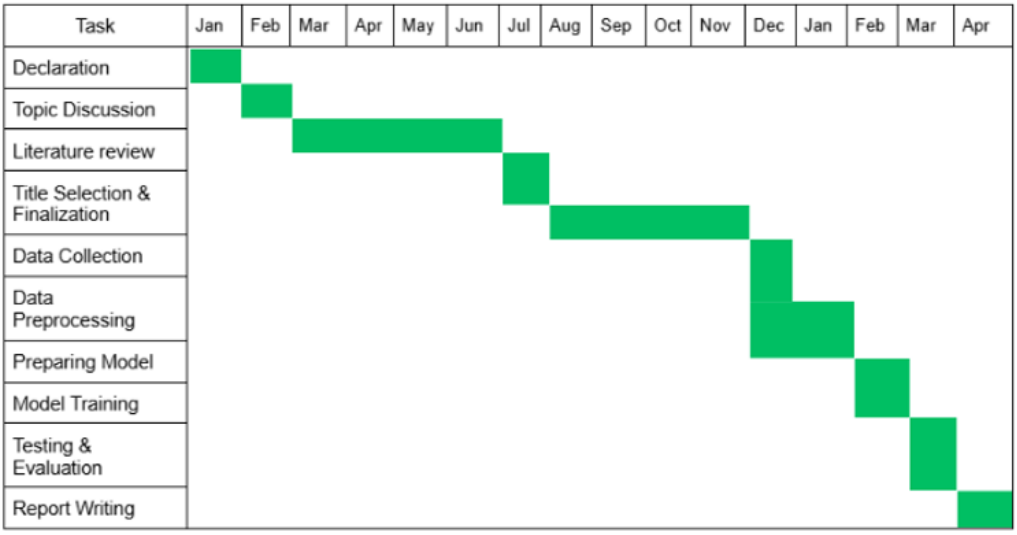


Figure 3.3: Timeline of the project.

The project timetable is methodically designed to ensure a smooth execution and early conclusion. It begins in January with the Introduction phase and the initial supervisor meeting, during which basic goals and scope are determined. Topic Discussion, which takes place between February and March, focuses on refining the project's direction and objectives.

The Literature Review phase, which runs from March to April, enables a thorough grasp of current research on Bengali Word Sense Disambiguation (WSD) and associated methodologies. Following that, May is dedicated to title selection and finalization, which confirms the project's focus before the Title Defense in June, when the proposal is formally authorized.

Dataset collection takes conducted between June and July, and the essential Bengali language datasets for WSD are acquired and prepared for model training. From August to November, frequent Follow-Ups maintain continuous progress, address any issues that occur, and ensure consistency with project objectives.

Model testing takes place in October and November and includes assessment metrics, performance analysis, and comparisons to current benchmarks. Finally, in December, the project ends with the Final Defense, which presents and evaluates the project's outcomes and contributions.

This timeline provides for concentrated, controlled development at each phase, ensuring that all activities are done on time while also allocating enough time for iteration and refinement to assure the project's success.

Chapter 4

Design and Development

Developing a robust Word Sense Disambiguation (WSD) system for the Bangla language involves leveraging recent advances in transformer-based language models while addressing the unique challenges posed by the language's rich morphology and semantic ambiguity. To maximize performance and explore model effectiveness, this work employs two state-of-the-art pretrained Bangla BERT variants: **csebuetnlp/banglabert** and **sagorsarker/bangla-bert-base**. These models serve as the foundational encoders that provide contextualized word embeddings essential for distinguishing among multiple senses of polysemous words.

The overall design adopts a modular framework that facilitates the integration of both models within the same pipeline, allowing for experimentation with different architectures and comparison of their impact on disambiguation accuracy. This modularity also supports token-level enhancements, such as the insertion of special markers around the ambiguous target word, which helps the model focus attention on the relevant part of the input sentence.

Data collection and integration constitute a critical phase, involving the curation of a comprehensive dataset featuring carefully annotated polysemous Bangla words with multiple senses, supported by context-rich example sentences. This dataset underpins the supervised training of both BERT models, enabling them to learn nuanced semantic distinctions.

The model design centers on fine-tuning each pretrained BERT encoder with a classification head that projects the contextualized '[CLS]' token embedding into a probability distribution over the possible senses via PyTorch argmax function. This design choice ensures that both models can be trained and evaluated under

comparable settings, facilitating an objective assessment of their strengths and limitations.

Subsequent sections provide a detailed exposition of the system's architectural layout, data strategies, and model configurations, forming a comprehensive account of the design and development journey.

4.1 System Architecture Design Pipeline:

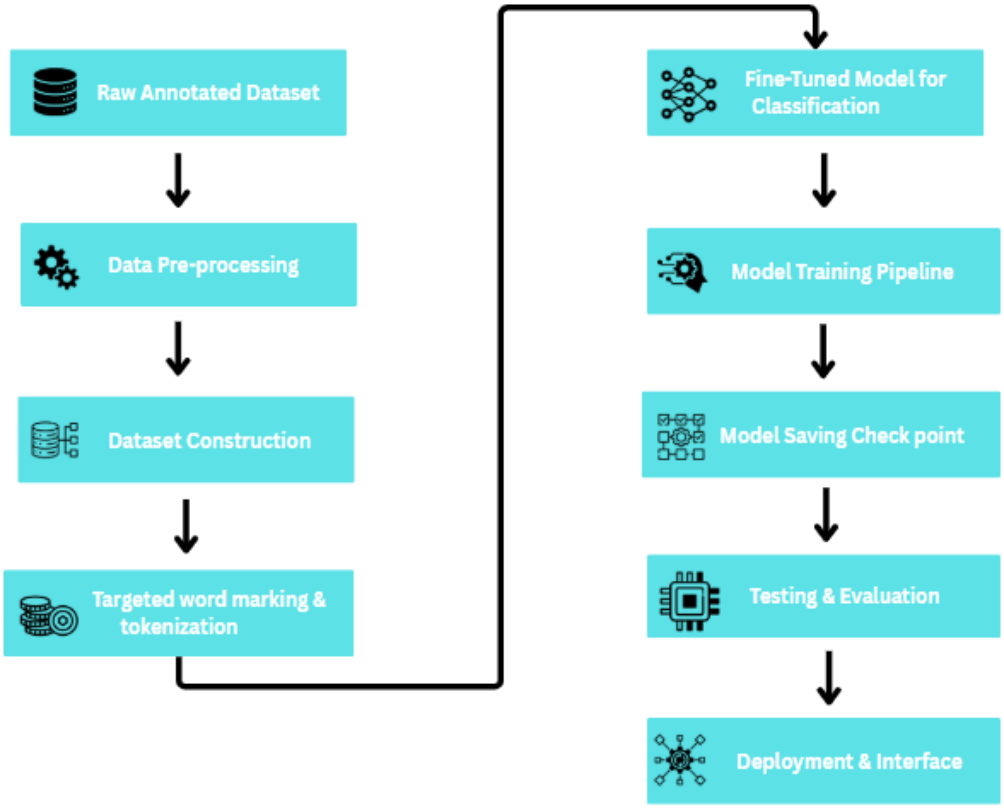


Figure 4.1: proposed system architecture

The proposed Word Sense Disambiguation (WSD) system for Bangla is organized into a modular pipeline comprising several key components, each handling a specific responsibility in the overall workflow. The architecture ensures seamless transition from raw annotated data to final model inference. The following subsections describe the structure and functionality of each part in detail.

4.1.1 Raw Annotated Dataset (JSON Format)

The system begins with a manually annotated dataset stored in JSON format, where each entry corresponds to a Bangla polysemous word. The structure of each entry consists of a 'word' field that identifies the ambiguous word, followed by a 'senses' array. Each item within the 'senses' array contains a 'sense' label, which specifies a particular meaning of the word, along with an 'examples' array that includes multiple sentences demonstrating how that sense is used in natural language contexts. This format allows for clear alignment between each word's possible meanings and their real-world usage, forming a reliable foundation for training a word sense disambiguation model.

4.1.2 Data Preprocessing & Label Encoding

The data preprocessing in the project is handled by a dedicated module named "BengaliWSDDataProcessor", which is responsible for preparing the dataset for training and evaluation. The first step in this process is "flattening", where the originally nested data structure—organized as words mapped to multiple senses, each with several example sentences—is transformed into a flat format. In this flattened representation, each record contains a single sentence, the target ambiguous word, and the correct sense label. Following this, the processor performs "label encoding" to facilitate model training. It compiles all unique sense labels from the dataset and creates two essential mappings: "label2id", which assigns a unique integer ID to each sense label, and "id2label", which reverses this mapping to retrieve the original sense label from its corresponding ID. This structured approach ensures the data is in a format suitable for machine learning models to process effectively.

4.1.3 Dataset Construction

After preprocessing, the flattened and encoded data is loaded into a HuggingFace-compatible 'Dataset' object, which facilitates streamlined handling for model training and evaluation. This dataset is structured with the following key fields: 'sentence', which contains the full sentence including the ambiguous word; 'target_word', specifying the word whose sense needs to be disambiguated; 'sense', representing the textual form of the word's intended meaning and 'label', the corresponding integer class ID assigned during label encoding. This standardized format not only

ensures consistency across the data but also enables efficient processing, batching, and integration with modern NLP training pipelines.

4.1.4 Target Word Marking & Tokenization

To help the model effectively identify and focus on the word that requires disambiguation, custom marker tokens '[TGT]' and '[/TGT]' are inserted around the target word** within each sentence. This annotated version of the sentence is then passed through a tokenizer compatible with the 'BanglaBERT model', ensuring the input format aligns with the model's expectations. The tokenization process produces three key components: 'input_ids', which are the numerical token IDs representing the sentence; 'attention_mask', which differentiates meaningful tokens from padding and 'label', which is the integer class ID indicating the correct sense of the target word. The use of these marker tokens is crucial, as it explicitly highlights the word of interest, thereby guiding the model's attention during training and improving the effectiveness of the disambiguation process.

4.1.5 Model: Fine-Tuned BanglaBERT for Classification

The tokenized inputs are fed into a fine-tuned transformer-based model configured for sequence classification. For experimentation, two pretrained Bangla BERT models are utilized: 'csebuetnlp/banglabert' and 'sagorsarker/bangla-bert-base'. To adapt these models for the word sense disambiguation task, a classification head is appended, consisting of a linear layer followed by a softmax activation function. This setup allows the model to generate a probability distribution over all possible sense classes for a given input sentence. Furthermore, to ensure compatibility with the custom preprocessing, the model's token embeddings are resized** to include the newly introduced special tokens '[TGT]' and '[/TGT]'. This adjustment enables the model to properly interpret and leverage the markers that highlight the disambiguation target, thereby enhancing its performance on the task.

4.1.6 Model Training Pipeline

Training is carried out using the HuggingFace Trainer API, which offers an efficient and standardized framework for fine-tuning transformer-based models. The model optimizes the CrossEntropyLoss function, appropriate for the multi-class

classification task of word sense disambiguation. For optimization, the AdamW optimizer is utilized due to its proven effectiveness in training large pretrained language models. Throughout the training process, key evaluation metrics including Accuracy, Precision, Recall, and F1-score are monitored to provide a comprehensive assessment of the model's performance. Training runs with a learning rate of 3e-5, over 15 epochs, and a batch size of 32. After each epoch, the model's performance is evaluated on the reserved 10% validation set, and only the checkpoint with the best validation performance is retained. This strategy ensures that the final selected model generalizes well and avoids overfitting.

4.1.7 Model Saving & Checkpointing

Upon completion of training, the final model along with its tokenizer are saved to the output directory '[Model Name]_bangla_wsd_model'. Throughout the training process, the framework manages intermediate checkpoints automatically, retaining only the best-performing version based on evaluation loss. This ensures efficient storage use while preserving the most effective model for deployment or further use.

4.1.8 Testing & Evaluation:

For evaluation, both the validation and test sets are used to assess the model's performance comprehensively. Key quantitative metrics such as F1-score, Precision, and Accuracy are calculated to measure classification effectiveness. Additionally, a confusion matrix is generated to visualize the distribution of correct and incorrect predictions across all sense classes, providing deeper insights into specific areas where the model performs well or struggles. To aid interpretability, the confusion matrix is displayed as a heatmap, highlighting patterns of misclassification. Furthermore, training progress is tracked through graphical plots of accuracy and loss over epochs, generated for both validation and test phases. These visualizations help monitor the model's learning behavior and diagnose potential issues like overfitting or underfitting, ensuring a robust and reliable evaluation.

4.1.9 Deployment and Inference:

In the deployment phase, the fine-tuned BanglaBERT model is utilized to perform word sense disambiguation on new, unseen sentences. The ambiguous target word within each sentence is clearly marked using special tokens [TGT] and [/TGT] to maintain consistency with the training process and help the model focus on the relevant word. These marked sentences are tokenized using the same tokenizer applied during training, ensuring proper input formatting for the model. The processed input is then fed into the model, which generates probability distributions over the possible word senses through a softmax layer. The sense with the highest predicted probability is selected as the final output. This approach allows the model to generalize well to diverse contexts and be effectively applied in real-world scenarios such as natural language understanding, machine translation, and information retrieval tasks involving Bangla text. The deployment design emphasizes both accuracy and efficiency, enabling practical use of the model for real-time inference.

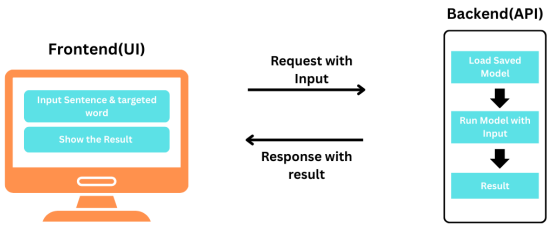


Figure 4.2: Model Prediction Interface

4.2 Dataset Integration

The dataset used in this system was manually constructed to support Bangla Word Sense Disambiguation. Due to the scarcity of high-quality, publicly available datasets for Bangla, a custom annotated dataset was developed in JSON format. Each entry in this dataset represents a Bangla polysemous word and contains a list of senses, with each sense associated with several example sentences demonstrating its use in different contexts. This structure allows the model to learn how context influences the meaning of an ambiguous word. The dataset is stored as a JSON file and is ingested into the system using the BengaliWSDDataProcessor class, as defined in the code. This processor reads the file and performs necessary preprocessing. It flattens the nested structure—extracting each sentence-sense pair as a separate training instance—and constructs a unified list of labeled examples. During this process, it also generates label mappings (label2id and id2label) that convert textual sense labels into numerical class IDs required for model training and prediction.Once processed, the data is formatted into a HuggingFace Dataset object. Each entry contains the sentence, the ambiguous word (target_word), the textual sense label (sense), and its corresponding class ID (label). This dataset is then split into training, validation, and testing subsets (80%, 10%, 10% respectively) to support both model development and evaluation phases. The integration ensures that the dataset seamlessly flows into the rest of the pipeline, maintaining compatibility with tokenization, model training, and evaluation procedures defined in the system.

4.3 Model Design

The design of the model is centered around the effective use of pretrained transformer-based language models, specifically BanglaBERT, to perform word sense disambiguation (WSD) in Bangla text. The architecture carefully highlights the target ambiguous word, processes the input through deep contextual encoding, and classifies the sense using a learned classification layer. Each step in the pipeline plays a crucial role in capturing contextual nuances and accurately predicting the intended meaning of polysemous words.

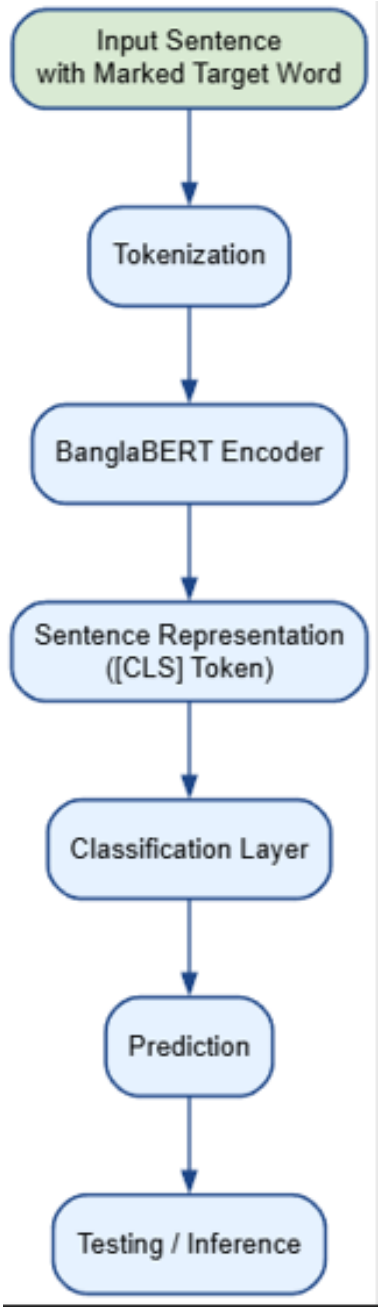


Figure 4.3: proposed model architecture

4.3.1 Input Sentence Preparation:

Each input sentence contains a target polysemous word that needs disambiguation. To explicitly inform the model about which word to focus on, the target word is enclosed within special markers [TGT] and [/TGT]. This marking guides the model

to pay special attention to this word's context while processing the sentence.

4.3.2 Tokenization:

The prepared sentence is tokenized using the tokenizer associated with the models. This tokenizer converts the sentence into a sequence of token IDs, which are numerical representations of subword units. Special tokens [CLS] (start of sentence) and [SEP] (end of sentence) are added to conform with the transformer input format. Additionally, an attention mask is generated to indicate which tokens should be attended to by the model during encoding, ensuring padded tokens are ignored.

4.3.3 Contextual Encoding:

The tokenized input is passed through the pretrained encoder, which consists of 12 transformer layers. Each layer applies self-attention mechanisms and feedforward neural networks to generate deep contextualized embeddings. The output is a sequence of hidden states—each corresponding to a token in the input—that encapsulate semantic and syntactic information from the entire sentence.

4.3.4 [CLS] Token Embedding Extraction:

From the sequence of hidden states, the embedding corresponding to the [CLS] token is extracted. This vector, with a fixed dimensionality of 768, is designed to represent a holistic summary of the sentence's meaning, including the context of the target word. It serves as the input feature vector for downstream classification.

4.3.5 Classification Head:

The extracted [CLS] embedding is fed into a fully connected linear layer which acts as a classification head. This layer transforms the embedding into logits corresponding to each possible sense class for the target word. A softmax function is applied to convert these logits into probabilities, yielding a distribution over the senses and allowing the model to output the most probable meaning.

4.3.6 Training and Evaluation Procedure:

During training, the model optimizes the CrossEntropyLoss between the predicted sense distribution and the true sense label, enabling it to learn discriminative patterns. In the evaluation phase, predictions are made by selecting the class with the highest probability (argmax). Performance is quantified through multiple metrics—Accuracy, Precision, Recall, and F1-score—to provide a comprehensive assessment of the model's classification capability.

4.3.7 Testing and Inference:

For unseen sentences during testing or real-world deployment, the same input preparation and processing pipeline is applied. The fine-tuned model predicts the most appropriate sense for the target word based on its learned knowledge, enabling effective word sense disambiguation in practical applications.

4.4 Adopted Tools and Techniques:

4.4.1 Software Requirements:

To ensure the effective operation of the system, the following software components are required:

Python 3.x: The project is built on Python 3.x, chosen for its robust ecosystem in machine learning, including support for libraries like TensorFlow, PyTorch, and Scikit-learn.

Machine Learning Libraries: TensorFlow, Keras, and Scikit-learn are essential for developing, training, and evaluating the models, including data preprocessing and augmentation tasks. These libraries support the creation of deep learning models and handle non-image medical data.

Version Control System: Git will be used for source code management, facilitating team collaboration.

Integrated Development Environment (IDE): Visual Studio Code (VS Code) are selected for efficient development, providing environments to manage dependencies and enhance the coding and debugging workflow.

Chapter 5

Result analysis

5.1 Model performance:

5.1.1 csebuetnlp/BanglaBERT Result:



Figure 5.1: Accuracy Plot.

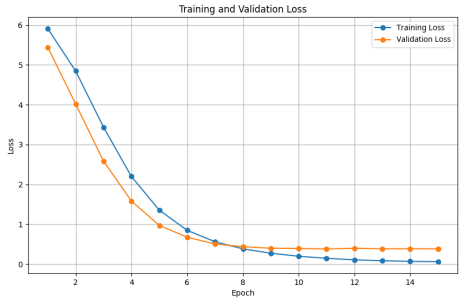


Figure 5.2: Loss Plot.

The training versus validation accuracy plot of the model csebuetnlp/BanglaBERT demonstrates a consistent and effective learning trend across epochs. In the early stages (up to around epoch 5–6), both training and validation accuracy increase rapidly, with the validation accuracy even rising slightly faster at first, which indicates that the model is quickly able to capture meaningful language representations. As training progresses, the two curves remain close to each other until about epoch 8, suggesting that the model is generalizing well without signs of underfitting. Beyond this point, the training accuracy continues to improve, eventually reaching around 98%, while the validation accuracy plateaus around 88–89%. This divergence shows a mild degree of overfitting, where the model becomes increasingly

specialized to the training data without further gains on unseen data. However, the gap is not severe, and the stable validation curve indicates that BanglaBERT achieves strong generalization performance for the task. Overall, the plot reflects that BanglaBERT is a powerful model for Bangla language understanding, achieving high accuracy while maintaining reliable generalization.

The training and validation loss curves of the csebuetnlp/BanglaBERT model demonstrate that the model rapidly learns useful representations in the initial epochs, as both losses decrease sharply from around 6 to below 1. In the mid-training phase, the two curves converge, with validation loss reaching its minimum of approximately 0.35–0.4 around epoch 8, while training loss continues to decline steadily. In the later epochs, training loss reduces further towards nearly 0.05, whereas validation loss plateaus, indicating that the model has effectively converged and that additional training mainly benefits memorization of training data rather than improving generalization. The small gap between training and validation loss suggests only mild overfitting, which aligns with the accuracy plot, confirming that BanglaBERT achieves high performance with stable validation behavior.

For illustration purposes, we selected 8 random classes out of 440 total classes to present the heatmap and confusion matrix for classification analysis. The confusion matrix illustrates the performance of the BanglaBERT model on 8 randomly selected classes from a total of 440. Each row represents the true labels and each column the predicted labels, and the results show that all non-zero values lie perfectly on the diagonal (e.g., 3, 2, 4, 4, 4, 8, 4, 4). This indicates that the model correctly classified every test instance within this subset, achieving 100% accuracy, precision, and recall for these classes. Although the evaluation is based on a relatively small number of samples per class, the clear diagonal structure highlights the strong discriminative ability of the model and reflects its promising effectiveness when scaled to the complete set of 440 classes.

The per-class precision, recall, and F1-score heatmap provides a detailed breakdown of the model. The results demonstrate a very strong overall performance, with precision scores of 1.00 for all classes, indicating that whenever the model made a prediction, it was always correct and did not generate any false positives. Recall values also remain consistently high, reaching 1.00 for nearly all classes, which shows that the model successfully identified all relevant samples for those categories. However, one class (“ ”) exhibits a slightly lower recall of 0.67, meaning

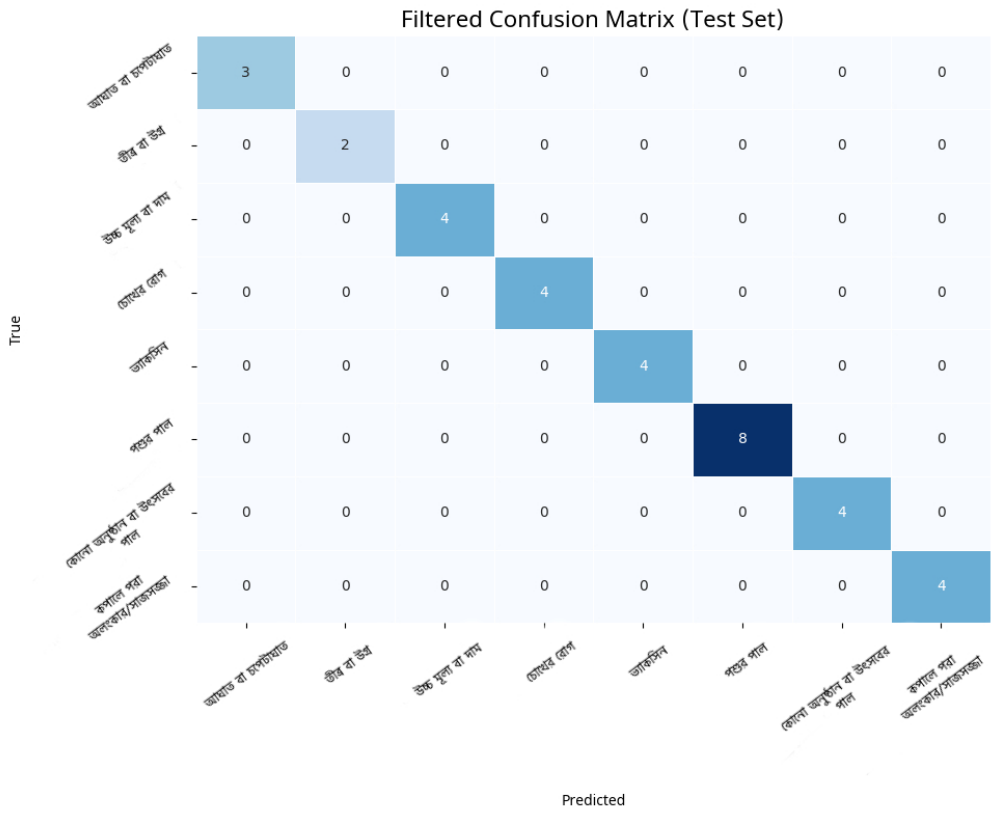


Figure 5.3: Confusion Matrix (Test Dataset).

the model missed one instance out of three. This drop in recall leads to a corresponding F1-score of 0.80 for that class, while the remaining classes maintain perfect F1-scores of 1.00. The support column highlights that each class contains only a small number of test instances, ranging from 3 to 8, with a total of 34 samples across all eight classes. This small sample size makes the evaluation sensitive to individual errors, as a single misclassification can significantly impact the metrics. Despite this limitation, the micro-averaged results (Precision = 1.00, Recall = 0.97, F1 = 0.99) clearly reflect the strong discriminative capability of the model in this subset. Together with the confusion matrix results, the heatmap emphasizes that BanglaBERT is highly effective in distinguishing between classes, producing consistent and reliable predictions, and shows promising potential when applied across the full set of 440 classes.



Figure 5.4: csebuetnlp Classification Report.

5.1.2 sagorsarker/bangla-bert-base Result:



Figure 5.5: Accuracy Plot.

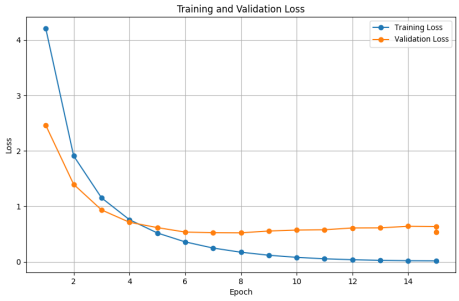


Figure 5.6: Loss Plot.

The training vs. validation accuracy plot illustrates the learning behavior of the sagorsarker/bangla-bert-base model over 15 epochs. Training accuracy (blue line) starts relatively low but rises consistently, reaching almost perfect accuracy (1.0) by the final epochs, which shows that the model fits the training data very well. Validation accuracy (orange line) also improves rapidly during the first few epochs and stabilizes around 0.80–0.82, with only minor fluctuations afterwards.

The widening gap between training and validation accuracy highlights a degree of overfitting, where the model continues to learn patterns from the training data but does not translate the same improvement to unseen validation data. Despite this gap, the model maintains strong validation performance, indicating it has learned meaningful representations for the task.

The training and validation loss plot of the sagorsarker/bangla-bert-base model over 15 epochs shows how the model minimizes errors during learning. Training loss (blue line) decreases sharply in the initial epochs and continues to decline steadily, approaching nearly zero by the final epochs, indicating that the model has fit the training data extremely well. Validation loss (orange line) also decreases rapidly at first and reaches a stable range around 0.5–0.7 from epoch 5 onward, but unlike training loss, it does not continue to improve. Instead, it begins to fluctuate slightly and even increases marginally in later epochs. This divergence between training and validation loss is a strong sign of overfitting, as the model continues to reduce error on the training set but fails to achieve the same level of improvement on unseen data. Nevertheless, the relatively low and stable validation loss suggests that the model still generalizes reasonably well.

The confusion matrix of the sagorsarker/bangla-bert-base model on 8 randomly selected classes out of the full 440 provides several key insights into the model's classification performance. A clear diagonal dominance can be observed, where the majority of samples fall into the correct prediction cells, showing that the model is able to correctly distinguish between most of the chosen categories. Specifically, the model correctly classified 3, 3, 4, 4, 4, 8, 4, and 3 instances for the respective classes, which reflects very high accuracy at the class level. This indicates that, in most cases, the model demonstrates strong discriminative power and robust understanding of linguistic variations across the sampled labels. A notable observation is that misclassification is extremely limited, appearing only in one class where the model correctly predicted 2 out of 4 instances, leading to slightly reduced recall for that class. Despite this minor weakness, the absence of widespread off-diagonal entries indicates that the model rarely confuses one class with another, suggesting reliable separation among these categories. The matrix also highlights that even with relatively small sample sizes per class, the model maintains consistent accuracy, which speaks to its effectiveness in learning meaningful representations. Overall, this confusion matrix confirms that sagorsarker/bangla-bert-base achieves strong

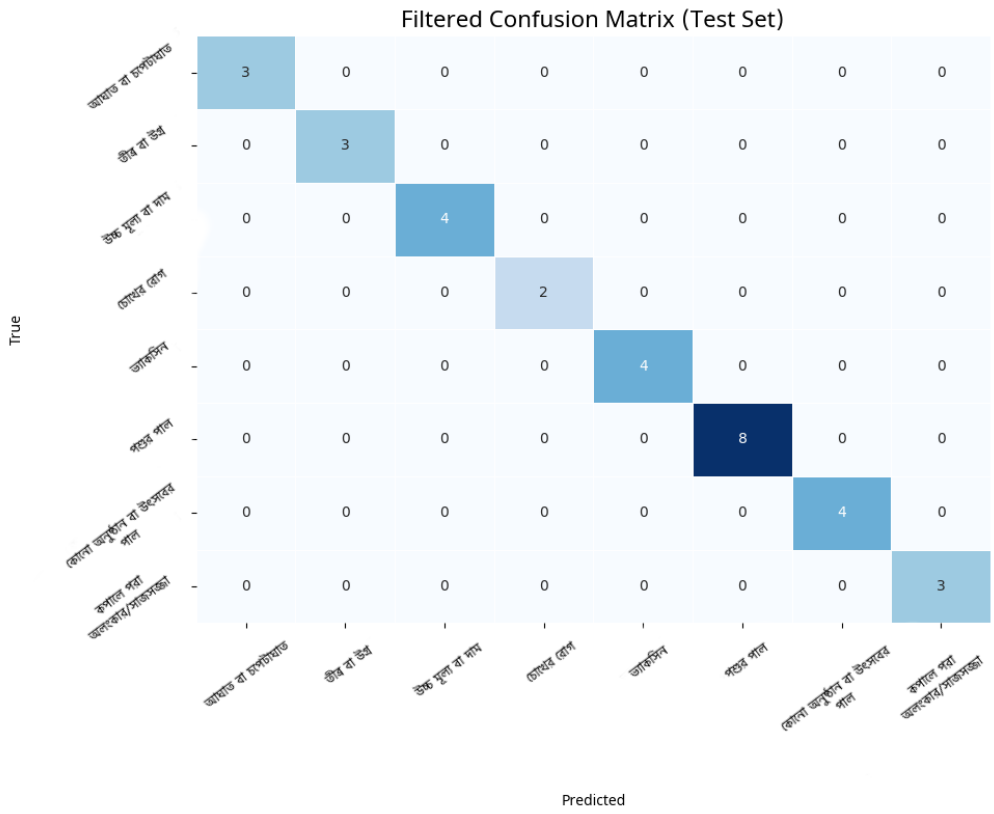


Figure 5.7: Confusion Matrix (Test Dataset).

and stable performance across the subset of classes, with near-perfect results for most and only a minimal performance gap for one, reinforcing its reliability as a multi-class classification model.

The heatmap analysis of the ‘sagorsarker/bangla-bert-base‘ model across 8 randomly selected classes out of 440 highlights that the model performs with very high precision across almost all classes (mostly 1.00), which means it very rarely predicts incorrect labels for these categories. However, recall shows more variability, ranging from 0.50 to 1.00, where the class ‘ ‘ stands out with a lower recall of 0.50, indicating that the model misses about half of its true instances, resulting in a reduced F1-score of 0.67 compared to other classes that maintain perfect F1 = 1.00. This suggests that while the model is highly confident in its predictions, it occasionally fails to capture all true cases for certain classes. The support values in this subset are relatively small and uneven, ranging from 3 to 8 samples, with ‘ ‘ being the most frequent class (support = 8), which may also influence the stability



Figure 5.8: sagorsarker Classification Report.

of results. Despite the limited sample size, the model shows strong robustness, with most classes performing exceptionally well. At the micro-average level, the model achieves precision = 1.00, recall = 0.91, F1 = 0.95, with a total support of 34, underscoring its overall strength in precision, minor weaknesses in recall, and reliable consistency across diverse categories even in a constrained evaluation setting.

5.2 Model Comparison of csebuetnlp/banglabert and sagorsarker/bangla-bert-base:

- **Training Stability:**
  Both models demonstrated stable training and validation behavior. csebuetnlp/BanglaBERT exhibited slightly higher training loss but stable convergence. sagorsarker/bangla-bert-base showed lower training loss and a comparable validation trend, indicating effective generalization in both cases.

- **Accuracy:**
  csebuetnlp/BanglaBERT: 88.60%
  sagorsarker/bangla-bert-base: 83.12%
  csebuetnlp/BanglaBERT achieved the highest overall accuracy.

- **Precision, Recall, F1-Score:**
  csebuetnlp/BanglaBERT: Precision 89.82%, Recall 88.93%, F1-Score 89.38%
  sagorsarker/bangla-bert-base: Precision 84.14%, Recall 83.12%, F1-Score 83.64%
  csebuetnlp/BanglaBERT consistently outperformed sagorsarker/bangla-bert-base across all macro metrics.

- **Loss Values:**
  csebuetnlp/BanglaBERT: Training Loss 0.086, Validation Loss 0.341
  sagorsarker/bangla-bert-base: Training Loss 0.016, Validation Loss 0.635
  While sagorsarker/bangla-bert-base had lower training loss, csebuetnlp/BanglaBERT showed stronger validation performance.

- **Observations:**
  The evaluation shows that both models learn effectively and generalize well on Bangla text classification. Training and validation accuracy trends indicate stable learning with only mild overfitting, while loss curves confirm convergence. Confusion matrix and heatmap analyses reveal high precision and strong discriminative ability across most classes, with only minor drops in recall for a few categories. Overall, the models demonstrate robust and reliable performance, capturing meaningful patterns in the data while maintaining consistent predictions across diverse classes.

Chapter 6

Conclusions

This study has successfully demonstrated a supervised deep learning approach for Bengali Word Sense Disambiguation (WSD) using state-of-the-art transformer-based language models. Specifically, the comparative evaluation of two prominent pre-trained models—csebuetnlp/banglabert and sagorsarker/bangla-bert-base—revealed that BanglaBERT achieved the highest performance across all major evaluation metrics. This work contributes significantly to the Bengali Natural Language Processing (NLP) field, particularly in resource-scarce domains like WSD, where annotated data and robust models are limited.

The csebuetnlp/BanglaBERT model attained an accuracy of 88.60% on test dataset, outperforming its counterparts while exhibiting consistent precision, recall, and F1-scores across both frequent and nuanced senses of ambiguous words. This demonstrates the model's strong generalization capabilities and robustness in handling linguistic variations inherent in Bengali. Furthermore, classification reports and confusion matrices indicated improved class separation, fewer false positives, and better interpretability—essential characteristics for deploying real-world NLP systems in low-resource languages.

The design and preprocessing of a high-quality, manually curated Bengali WSD dataset played a vital role in this achievement. By ensuring that each ambiguous word was annotated with multiple sense-labeled sentences drawn from diverse literary and colloquial sources, the dataset helped in capturing a wide semantic spectrum. The data preparation process also addressed class imbalance issues using stratified sampling and weighted loss functions, leading to improved model learning.

While the results are promising, challenges such as sense granularity ambiguity, limited training data for rare senses, and linguistic overlaps remain. Additionally, evaluating models beyond traditional metrics—using human judgment or downstream task integration—could provide richer insights. Nonetheless, the methodology and results provide a foundation for advancing supervised Bengali WSD research and building practical applications such as intelligent search engines, grammar checkers, and machine translation systems for Bengali.

6.1 Social, Legal, Ethical, and Environmental Issues

The deployment of a supervised Bengali WSD model based on BERT-style architectures raises several important considerations beyond technical achievements. These include the social impact on language accessibility, legal aspects related to data usage, ethical challenges in AI fairness, and environmental implications of model training.

6.1.1 Social Impact

This research can significantly enhance linguistic tools for Bengali speakers and democratize access to high-quality language technologies in underserved regions. It supports:

- Preservation of Bengali linguistic richness, particularly in educational and cultural tools.
- Enhanced accessibility for digital platforms, such as voice assistants, spell checkers, and learning aids.
- Support for language equity, giving Bengali the same technological advantages available to high-resource languages.

6.1.2 Legal Considerations

Legal challenges are primarily associated with the collection and annotation of textual data, particularly in preserving:

- Copyright compliance when extracting texts from literary sources.

- Transparency in data sourcing, ensuring that usage rights and ethical standards are upheld.
- Compliance with regional data protection laws, especially if user-contributed content is involved in future dataset expansion.

6.1.3 Ethical Implications

Ethical considerations are paramount in developing AI models for linguistically diverse populations. This includes:

- Bias mitigation in model predictions to ensure no sociolect or dialect is unfairly disadvantaged.
- Transparency and explainability in model outputs, especially in educational or decision-support applications.
- Respect for linguistic and cultural context, avoiding reductionist interpretations of rich semantic nuances.

6.1.4 Environmental Sustainability

Training transformer models incurs a considerable computational cost. Efforts to mitigate this include:

- Efficient fine-tuning strategies (e.g., parameter-efficient tuning, early stopping) to reduce training overhead.
- Cloud computing with renewable energy resources when training at scale.
- Reuse and open sharing of models, reducing the need to train large models from scratch.

6.2 Brief Summary

This work presented a comprehensive, supervised Bengali Word Sense Disambiguation system using fine-tuned BERT-based models on a custom-annotated dataset. The findings demonstrate the feasibility and effectiveness of using transformer architectures in a linguistically complex, low-resource setting. By leveraging

`BanglaBERT`, the system achieved high accuracy and robust generalization, laying the groundwork for future advancements in Bengali NLP. This project not only contributes to the computational linguistics community but also supports broader social, educational, and technological objectives for Bengali language development.

6.3 Future Works

The Bengali Word Sense Disambiguation (WSD) system developed in this study establishes a strong foundation for advancing semantic understanding in Bengali NLP, with multiple avenues to extend its impact. A key direction for future work is the systematic expansion of the dataset. This involves not only increasing the number of contextually annotated sentences per sense 200 or more sentences for common polysemous words and more for highly ambiguous ones—but also incorporating examples from colloquial, dialectal, literary, and technical domains. Such expansion would enhance the model's ability to generalize across the rich linguistic landscape of Bengali and capture subtle semantic distinctions.

Another critical aspect is enhancing the vocabulary coverage of the WSD system. Extending the system to handle a larger set of polysemous words, including domain-specific, technical, and less frequent words, will make it more versatile and practically useful. This improvement will allow the system to address a wider variety of real-world text and applications, from literature analysis to specialized technical documents.

Evaluating the dataset and system with multiple architectures, such as LSTM-based models, BERT variants, mBERT, and XLM-RoBERTa, can provide insights into the model's adaptability and highlight potential limitations in capturing semantic nuances. Integration into downstream applications, including machine translation for sense-specific accuracy, question-answering for precise semantic matching, and writing-assistant tools for students and professionals, will broaden practical utility. Cross-lingual and zero-shot learning with multilingual transformers like XLM-RoBERTa can extend WSD to under-resourced Bengali dialects and related South Asian languages, promoting linguistic inclusivity.

Finally, implementing adaptive learning mechanisms can allow the system to continuously improve over time by incorporating user feedback and tracking evolving language patterns. Leveraging cloud-based computing resources can support

this process, enabling real-time updates and enhancements without requiring high computational power on local devices. Pursuing these directions—dataset expansion, vocabulary enhancement, architectural evaluation, cross-lingual learning, and adaptive cloud-based learning—can drive scalable, interpretable, and high-impact semantic systems, advancing Bengali NLP and supporting broader South Asian language processing.

Chapter 7

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